

Spectroscopic Follow-up Orchestration Layer

Design & Implementation Review for the MAGNETS Collaboration

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Reflects the post-review implementation

Abstract

This document describes the design and current implementation of the orchestration layer for automated spectroscopic follow-up of transients discovered by the RubinAlerts pipeline, targeting LLAMAS on Magellan/Baade for the MAGNETS collaboration. It supersedes the April 2026 draft and incorporates a full design review (18 findings across ingestion, merit scoring, prioritization, scheduling, time accounting, and transparency) together with two subsequently-added capabilities: a cross-night *per-target integration ledger* and *per-program light-curve phase preferences*. The guiding constraints throughout are the PI mandates: **no wasted time on sky** and **transparent, defensible prioritization**. The system is exercised by 96 offline unit tests (no live broker/database calls).

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1 Overview

The orchestrator converts ranked candidates into a LLAMAS observing plan with multi-program time accounting. It ingests targets from two streams:

1. **Alert-driven** — the RubinAlerts pipeline’s `candidates.csv` (merit-ranked SN candidates aggregated across brokers); and
2. **Manual** — a CSV of PI-entered targets (the eventual Google-Sheet front end is future work; see §11).

The system is **LLAMAS-only** (Magellan/Baade; LDSS3 materials in `ref/` are reference only). The pipeline of modules is:

`normalize` → `prioritizer` → `planner` → `accounting` + `target_ledger` → `output/reporting`

2 Alert ingestion and liveness

2.1 Brokers

Candidates are aggregated from independent brokers, each with different classifiers, then cross-matched. ZTF data enters only via ALerCE-ZTF and ANTARES; there is no standalone ZTF client.

Broker	Survey	Role
Fink	Rubin LSST	Primary LSST discovery + SN classification
ANTARES	ZTF + Rubin	Galaxy-centric cone search; P(Ia) is a heuristic <i>proxy</i> (not ML)
ALerCE-ZTF	ZTF	BHRF forced-photometry classifier
ALerCE-LSST	Rubin LSST	Stamp classifier (cone-search API ignores coordinates; queried globally then filtered locally)
ATLAS	—	Forced-photometry <i>enrichment</i> for bright (< 20 mag) candidates
TNS	—	Cross-match / duplicate & spectroscopic-confirmation check
Rubin RSP/TAP	Rubin LSST	Host-galaxy properties, authoritative photometry

2.2 Coverage and the Rubin downtime window

The southern Deep Drilling Fields (ELAIS-S1, EDfS_a/b, $\delta \lesssim -32^\circ$) lie below ZTF/ANTARES/ALerCE-DB coverage and are therefore **single-broker by construction**. This matters operationally: **Rubin LSST is offline 2026-08-09 through the end of August 2026**, during which the ZTF-fed brokers are the *primary* alert source and the southern DDFs lose discovery coverage entirely.

2.3 Liveness reporting (review finding R5)

Brokers previously swallowed errors and returned empty results, so a silent outage was indistinguishable from a quiet sky — unacceptable during the Rubin downtime. The aggregator now records, per broker, a status record `{queried, responded, n_returned, error}`, written to a

`broker_status.json` sidecar and rendered as a “Broker Status” block in the run report. A broker raising an exception is recorded as queried-but-unresponsive rather than crashing the run.

2.4 Cross-matching (R13)

Cross-broker de-duplication uses true angular separation (`SkyCoord.separation`) against a fixed radius, replacing the previous RA/Dec box test whose effective radius shrank with declination (worst in the southern fields).

3 Merit scoring (alert pipeline)

The alert pipeline ranks candidates by a *multiplicative* merit, so a candidate must score well on *every* axis:

$$\text{Merit} = w_{\text{time}} w_{\text{mag}} w_{\text{prob}} w_{\text{host}} w_{\text{ext}} w_{\text{broker}} w_{\text{moon}} (w_{\text{salt}} w_{\text{absmag}}). \quad (1)$$

- $w_{\text{time}} = \exp(-\Delta t^2/2\tau^2)$, $\tau = 10$ d — peaks at maximum light.
- $w_{\text{mag}} = \exp(-((m - m_{\text{opt}})/\sigma_m)^2)$, $m_{\text{opt}} \approx 20.5$ (Magellan S/N sweet spot).
- w_{prob} — **ML-derived P(Ia) only (R7)**. The uncalibrated ANTARES heuristic proxy is kept in a separate `antares_proxy_prob` column and excluded from the pooled probability; ANTARES contributes to detection/coverage, not to P(Ia).
- $w_{\text{host}}, w_{\text{ext}}$ — host-morphology and Galactic-extinction factors.
- w_{broker} — **coverage-aware (R6)**: $1 + 0.3 \text{clip}(\frac{n-1}{n_{\text{max}}-1}, 0, 1)$ where n_{max} is the number of brokers that *could* have seen the field given its declination. A single-broker southern field is no longer penalized relative to a single-broker equatorial field; full multi-broker coverage still earns up to +30%.
- w_{moon} — **now folded into the ranking merit (R1)**. Previously the moon penalty was computed during observability filtering but never multiplied into the ranked merit, while the report *claimed* it had been applied. Merit is now recomputed with the real moon penalty as a single source of truth, and `candidates.csv` (the orchestrator’s input) carries the moon-aware merit.

4 Prioritization (orchestrator)

4.1 Composite score (R2, R14)

Each target receives a composite score that is a *multiplicative science core* plus two *additive nudges*:

$$\text{science_term} = \text{SCIENCE_SCALE} \times s \times b \times p \times c \quad (2)$$

$$\text{observability_term} = \text{OBSERVABILITY_BONUS} \times o \quad (3)$$

$$\text{keyword_term} = \text{KEYWORD_SCALE} \times k \quad (4)$$

$$\text{total} = \text{science_term} + \text{observability_term} + \text{keyword_term} \quad (5)$$

with `SCIENCE_SCALE= 100`, `OBSERVABILITY_BONUS= 20`, `KEYWORD_SCALE= 50`. These constants, previously undocumented magic numbers, now carry rationale in code: the science core ($s, b, p, c \in [0, 1]$) is the dominant axis (a P1 at peak on a healthy budget ≈ 100 ; a P4 ≈ 20), while observability and keyword nudges reorder within or across adjacent tiers but cannot by themselves leapfrog a full priority tier. The phase factor is clamped to $[0, 1]$ so a malformed weight can never blow past the science core. Every target’s component **breakdown is persisted** (`score_breakdown.json`) and printed in the summary, so a PI can reconstruct any ranking by hand — where:

<i>s</i> science	P1–P4 → 1.0/0.7/0.4/0.2
<i>b</i> budget	phase-aware factor 1.0/0.5/0.1 (§6)
<i>p</i> phase	per-program phase preference (§4.2)
<i>c</i> completeness	per-target integration ledger factor (§7)
<i>o</i> observability	fraction of night above the airmass limit
<i>k</i> keyword adj.	signed boosts/demotions from target notes

4.2 Per-program light-curve phase preference

Motivation. Different programs want a SN at different phases. Cosmology / standardization science favors *peak* (best S/N, the canonical SALT epoch, and where luminosity-correlating spectral indicators are measured). Progenitor, circumstellar-material (“flash”), early-excess, and fast/exotic-transient science favor the *rise* (outer ejecta, narrow CSM lines that are gone by peak, and transients that evolve too fast to wait). A single peak-centered weight cannot serve both.

Implementation. The phase factor is now computed relative to a per-program preferred epoch:

$$p = \exp\left(-\frac{(\Delta t - \Delta t_{\text{pref}})^2}{2\tau^2}\right), \quad \Delta t_{\text{pref}} = \begin{cases} 0 & \text{preference } \mathbf{peak} \\ -7 \text{ d} & \text{preference } \mathbf{rising} \end{cases} \quad (6)$$

with $\tau = 10$ d. Each `ProgramAllocation` carries a `phase_preference` (`peak` default; e.g. Stubbs/Ia = `peak`, Villar/exotic = `rising` in the example allocations). This requires a *signed* Δt (days from peak; < 0 rising), threaded from the `delta_t` column of `candidates.csv` or from `peak_mjd` for manual targets. The offset $\Delta t_{\text{pref}} = -7$ d is a tunable config value; its exact choice is a science-policy decision for the collaboration. Note an alert-sourced target without a `delta_t` value falls back to the legacy symmetric weight (peak-meaningful only), since the sign is unrecoverable from w_{time} alone.

4.3 Priority labels (R8)

Merit maps to P1–P4 by *tonight’s quartiles*; these are **within-night relative** labels, not absolute science classes (a “P1” on a quiet night may be weaker than a “P4” on a rich one). The summary header states this caveat, and the < 4 -target degenerate case is guarded by rank-based assignment. Absolute, cross-night-calibrated thresholds were deliberately deferred until merit statistics across nights exist.

5 Scheduling

A greedy scheduler steps through the night, at each slot selecting the observable target maximizing

$$\text{score} - 10 \times \text{airmass} - \text{SLEW_PENALTY} \times \theta_{\text{slew}}, \quad (7)$$

where θ_{slew} is the angular distance from the previously scheduled target (`SLEW_PENALTY`= 0.5). The slew term (R9) is a modest tie-breaker — the IFU’s 1-minute overhead model assumes negligible slew, which is reasonable for clustered DDF targets but needed a guard against large cross-sky jumps. The scoring mode (`prioritizer` vs. `fallback/priority-only`) is stamped into the summary header (R18). Standard stars are placed at start and end *and* interleaved at a configurable mid-night cadence (`standard_interleave_hours`= 3.5; R10) for spectrophotometric calibration over a long night. Exposures are split into sub-exposures of at most `max_single_exposure_sec`= 900 s for cosmic-ray mitigation (R16, now config-driven).

6 Time accounting

Per-program budgets are defined per moon phase (dark/grey/bright) in `allocations.yaml`. Time is **charged on schedule** and trued-up post-night by `reconcile`. Key review fixes:

- **State-path consistency (R3)**. `run-nightly` and `reconcile` now derive the state file from the same `output_dir/time_accounting.json` convention (helper `_default_state_path`); previously `reconcile` defaulted to a different path, loaded empty state, and silently double-charged.
- **Science-time charging (R4)**. Charges `charged_minutes` (exposure + overhead) rather than the padded wall-clock window; gap-fill / end-of-night stretch is tracked as `padding_minutes` for transparency but never billed.
- **Phase-aware budget factor (R11)**. The 1.0/0.5/0.1 budget factor uses remaining hours *in the relevant moon phase* as a fraction of that phase’s allocation ($> 50\% \rightarrow 1.0$, $> 0 \rightarrow 0.5$, exhausted $\rightarrow 0.1$), instead of a global hour cliff.

7 Per-target integration ledger

A cross-night ledger (`target_ledger.json`, stored beside the accounting state) answers a question the per-program budgets cannot: *does this target already have enough integration time that we should stop re-observing it?* Without it, five consecutive nights could keep re-scheduling the same high-merit object.

7.1 Identity, completeness, and remaining-time scheduling

Targets are keyed by sky coordinate (matched within $2''$ angular separation, so a name change between nights — e.g. an internal alert ID later upgraded to a TNS/IAU name — is still recognized as the same object; names are tracked as aliases with the TNS name preferred as canonical). For each target:

- `completeness_fraction` = cumulative science seconds / required seconds, where the required exposure is *recomputed every night* from the current magnitude/redshift — so a fading SN correctly needs more time.
- A target is **satisfied** (excluded from scheduling) at fraction ≥ 0.95 or when the remaining time is below a worthwhile 15-minute block.
- The completeness factor folded into the score is $\max(0.15, 1 - \text{fraction})$, dropping to 0 when fully done. The 0.15 floor prevents the opposite failure mode — a target getting one short exposure and then being dropped forever — while the taper prevents endless re-picking.
- A partially-observed target is scheduled for only its *remaining* needed time.

7.2 Phase-split integration count

The cumulative count is **split by phase bucket** derived from the observed Δt : **rising** ($\Delta t < -5$ d), **peak** ($|\Delta t| \leq 5$ d), **declining** ($\Delta t > 5$ d), or **all** when Δt is unknown (`phase_bucket_window_days=5`). Completeness is judged against the bucket the SN is in *that night*, so a target that is “done at peak” can still be scheduled for needed rising-phase time, and vice versa. The ledger JSON format migrates old scalar records into the **all** bucket automatically.

7.3 Multi-group alert

When a single sky object is associated with more than one program — whether from two same-position requests on one night or from a program already recorded against that object on a prior night — the system emits an alert: a `logger.warning`, a “Multi-group Targets” section in the summary, and a `multi_group_alerts.json` sidecar with schema `{name, ra_deg, dec_deg, programs, phase_preferences, observed_phase, same_phase}`. The `same_phase` flag tells PIs whether the groups want the object at the same phase (one observation may serve both) or different phases (coordination / separate observations needed). Conflict *arbitration* is intentionally left to the PIs for now (see §11).

8 Exposure estimation

The orchestrator estimates LLAMAS exposure via a cascade: a redshift→time table from the proposal (now **linearly interpolated** to remove the previous hard cliffs, R12, clamped outside the tabulated range), falling back to magnitude scaling (mag 20 $\approx$ 45 min, 2.5× per magnitude), then a 45-minute default. The IFU overhead is $\sim$ 1 min (vs. $\sim$ 10 min for slit instruments). The orchestrator deliberately omits moon/airmass scaling here (the proposal table is the reference); this is a documented divergence from the alert-pipeline estimator, not an inconsistency.

9 CLI and outputs

```
python -m orchestrator plan          --date ... --targets t.csv --moon
    grey
python -m orchestrator run-nightly    --date ... --candidates c.csv \
    --allocations a.yaml --moon grey --output-dir out/ \
    [--target-ledger out/target_ledger.json]
python -m orchestrator reconcile      --program P --actual-hours 3.5 ...
python -m orchestrator reconcile-target --name "SN_2026abc" --actual-
    minutes 90 \
    [--phase peak] --date ...
python -m orchestrator ledger         --target-ledger out/target_ledger.
    json \
    [--show-satisfied]
```

Transparency artifacts written per run: the timeline, TCS catalog, and summary; plus `score_breakdown.json` (per-target score components), `time_accounting.json` (budget audit), `target_ledger.json` (per-phase integration), `broker_status.json` (ingestion liveness), and `multi_group_alerts.json`.

10 Testing

The system has 96 unit tests (`tests/`). A hard rule: **no test contacts a live broker, database, or API** — brokers are stubbed and all state uses temp-file JSON. Coverage spans merit (moon folding, coverage-aware broker bonus), ingestion (liveness status, ML-only probability, angular-separation dedup), accounting (state path, science-time charging, phase-aware budget factor), prioritization (normalized score, breakdown invariant, slew penalty, small-list guard), exposure interpolation, standards insertion, the integration ledger (completeness, phase splitting, migration), phase preferences, and multi-group alerts.

11 Open questions and future work

1. **Manual-request front end.** A Google-Sheet ingester (gsread + service account) is planned so PIs can enqueue targets directly; the CSV path is the current stand-in.
2. **Phase-preference tuning.** The `rising` offset (-7 d), τ , and the phase-bucket window (5 d) are science-policy values to be set with Chris and Ashley's group; per-program offsets (not just `peak/rising`) are a natural extension.
3. **Multi-group conflict arbitration.** Today we alert; a future option is to model an object wanted at two phases as two distinct requests, each with its own ledger bucket and required time.
4. **Absolute priority thresholds.** Replace within-night quartile P-labels with absolute merit thresholds once cross-night merit statistics are available.
5. **Multi-night look-ahead** scheduling, and surfacing broker liveness through the `supernova_monitor` pipeline path (currently only the `run_tonight` path).